LIVE DEMO & OPEN BUILD · OCTOBER 28, 2026

ARGUS-OS1

An Open-Source, AI-First Microscope for Longevity Science

A ~$3K microscope that sees, tracks and decides — entirely on hardware you own, no cloud.

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Open Q&A at the end · anyone building or benchmarking is welcome

THE WHY

The problem

Aging research is bottlenecked by instruments that lock your data in.

- Live-cell aging experiments run for **days** — and you want to watch the **same cells** the whole time.
- **Commercial systems** (Zeiss / Nikon / Leica class): closed optics, closed software, ~$100K+, data goes through vendor pipelines.
- **Open hardware** (OpenFlexure): fully open, ~$1K — but images on a fixed schedule; analysis happens later, off-instrument.
- **Cloud AI pipelines**: run heavy models, but raw biology leaves the building — every frame costs money and a network call.

The gap: nobody closed the loop on the instrument itself — acquire → segment → track → decide — with zero network dependency and every layer open.

THE MODEL SYSTEM

Why C. elegans

The ideal first target for an AI-first microscope: a fully mapped organism.

Feature	Value
Complete cell lineage	959 cells, every division axis known (Sulston & Horvitz 1977; Sulston et al. 1983)
Fixed division axes	A/P, D/V — “parallel vs perpendicular” is objectively defined
Fate of every cell	Known a priori — you can predict and check
Centriole fate map	~88% of cells eliminate centrioles in embryogenesis (Kalbfuss & Gönczy 2023)

Ground truth at every step — segmentation errors are verifiable against a published map.

THE SCIENCE

The hypothesis

Does a centriole's division history predict whether it lives or dies?

- **Pedigree Score** — a PCA composite of 5 division-geometry metrics: fraction of parallel divisions, mean 3D angle change, angle variance, orientation switches, cumulative angular path.
- **Question:** does geometry predict centriole fate beyond cell type, polarity and age?
- **Cleanest test — sister-cell pairs:** daughters share cell type, cytoplasm and lineage; the only systematic difference is which centriole each inherits.
- **Longevity link:** centriole homeostasis connects to longevity signalling converging on DAF-16 (insulin/IGF-1; heat-shock HSF-1 → PFD-6; Son et al. 2018) — a testable bridge between centrioles and lifespan.

“We ask whether a cell's history — written in its division angles — decides the fate of its centriole.”

THE BUILD

Hardware — open, ~$3K

Subsystem	Spec	Cost
Frame + stage	OpenFlexure v6.1.5, 3D-printed, Sangaboard motors	$800
Objective	40×/0.75 NA dry	$550
Camera	Raspberry Pi HQ (IMX477, C-mount)	$120
Compute	NVIDIA Jetson Orin NX (on-device inference)	$500
Climate	PID (Pico W) + SSR + PT1000 + HEPA H13	$346
Enclosure	acrylic incubator housing, fasteners	$160
Power / misc	12/24 V supplies, wiring, PCB	$320
Assembly & calibration	calibration target, test slides, spare fasteners	$200
TOTAL	full item-level BOM in repo (hardware/ARGUS-OS1_BOM.csv)	$2,996 (~$3K)

Everything printable, replaceable, nothing proprietary. Jetson (5–15 W) instead of a cloud GPU.

THE BRAIN

Edge AI pipeline — the loop is on the device

- **CellPose 2.0** — cell & centriole detection on-device (TensorRT, FP16).
- **Bayesian lineage tracker** — division tree frame by frame; every hypothesis carries uncertainty.
- **Biological scheduling** — the instrument decides what to image next from what it observes, not a fixed clock.
- **Age-related readouts** — division-rate dynamics, sister-pair comparisons, Hayflick kinetics in immortalized fibroblasts.

Core loop is 100% local

segment → track → schedule → log. Nothing depends on an external service.

Inspectable by design

Every decision logged with its input frames and model outputs — results are auditable against the raw record.

THE PRINCIPLE

Local compute — your frames never leave

- Raw images and derived features stay on **physically-owned hardware** — reproducible longevity measurement without shipping biology to a cloud.
- No account, no API fees, no bandwidth dependency — an experiment can run **unattended for days.**
- **DAIS — a verifiable AI layer:** every autonomous decision gets a **passport**, every intervention a **trace**, every knowledge claim a **provenance** — **PROPOSED ≠ FACT.**
- **Safety-critical steps** (firmware, stage control, graph traversal) stay deterministic; LLMs are confined to well-defined lanes.

AI-first bio on hardware you own — this is what the evening is about.

THE DIFFERENTIATOR

Phototoxicity + a closed box

Commercial systems are built for
hours–days

	Commercial	ARGUS-OS1→OS4
Observation	hours–days; population level	months, autonomous single-cell lineage
Phototoxicity	continuous light; not the driver (even brightfield kills — PMID 42393487)	minimal by architecture: LED + image-on-demand + label-free; light-sheet 10× lower; OET ~10^5× lower light intensity; dose audited per frame
Sterile in/out	none	transfer box: 2 doors, interlock, UV-C, HEPA
Manipulation in-box	none	glove ports + cable-driven robot arms (24/7) + micro-bots / micro-tweezers
Openness / cost	closed, $50–300K+	open, $3–5K (OS1) up the ladder

No vendor offers autonomous sample change.
ARGUS is designed past both the light wall and the service wall.

LIVE DEMO · NOW

What we'll show

1 · The agent loop

Local-first ARGUS agent (Rust, MIT) takes a request, plans, executes with local tools, writes the result. No cloud.

2 · DAIS passport & trace

A 24-hour experiment simulation: anomalies detected, decisions traced, knowledge with provenance.

3 · Tracking pipeline

Detection + lineage reconstruction on sample data (as available on stage).

4 · Repo walkthrough

BOM, docs, CI, open licenses — where you'd start building or benchmarking.

Honest scope: V6 physical assembly is in progress (hardware engineer, Tbilisi). Tonight: the software stack live + hardware state. We won't pretend otherwise.

HONEST LESSONS

What running AI-first bio on owned hardware teaches

- **The software loop is easier than the biology.** Segmentation/tracking on a Jetson is mostly solved; keeping a sample alive and in focus for days is not.
- Benchmarked against Zeiss Axio Observer 7: **focus-hold is the #1 gap** → prototyping fiducial autofocus on OpenFlexure (PMID 42274973).
- **Open hardware ≠ open pipeline.** OpenFlexure gives mechanics; closing the acquisition→analysis loop on-device was the real engineering work.
- **Local LLMs handle the boring 90%** — anomaly diagnosis, log mining, procedure lookup — and stay out of the safety-critical 10%.
- **A benchmark you can afford changes the conversation** — gaps are closable with open solutions, and we add what commercial systems lack: autonomous sample change (V9).

WHERE THIS GOES

Roadmap — one platform, four stages

Stage	Hardware	Budget	Goal
OS1 (now)	OpenFlexure + 40× dry + Jetson NX	~$3–5K	Local edge-AI platform, closed loop on device
OS2	60×/1.2 WI + sCMOS + microfluidics	~$70K HW	Core science: N=100 embryo pedigree experiment
OS3	+ light-sheet + fs-laser + tweezers	+$141K	Full platform + causality (ablation)
V9 (cross-stage)	Robot hands + shared local LLM brain	+$22K	24/7 self-servicing lab — autonomous sample change
OS4	Inverted + OET micro-manipulation	+OET module	NA 1.35–1.4, contact-free cell transport, micro-tweezers

OSC application submitted (Aug 2026) · NLnet Open Call submitted · Foresight RFP drafted for Oct 31 call.

CALL TO ACTION

Join the build

Build

Grab the BOM, print the frame, run the repo. Reproducible from a published bill of materials.

Benchmark

Run your own lineage-tracking data against ours — we want honest comparisons, not friendly ones.

Contribute

The gaps are known and public: focus-hold, LED engine, incubation — each an open engineering problem.

Repo: github.com/Georgia-Longevity-Alliance/ARGUS-OS1

Contact: jaba@longevity.ge

We built an instrument any lab can afford — now we need the community to make it right.

OPEN Q&A · ~10 MIN

Questions?

If the room goes quiet, try:

- Why centrioles for longevity science? → PFD-6/DAF-16 link, centriole homeostasis as a readout
- How is this different from OpenFlexure + offline CellPose? → the closed on-device loop
- What does “local LLM brain” mean in practice? → DAIS: deterministic core + confined LLM lanes
- How do we keep data trustworthy? → provenance, PROPOSED ≠ FACT, inspectable logs
- Fastest thing a newcomer can do this week? → run the repo, print the frame